INTERNATIONAL INSTITUTE OF INFORMATION TECHNOLOGY, BHUBANESWAR

Department of Computer Science & Engineering • CS201 Object-Oriented Programming

Embedded Doppler Visualizer

Real-Time Hardware-in-the-Loop Simulation of Acoustic Wave Compression and Supersonic Shockwaves

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Abstract — This report presents the design and implementation of the **Embedded Doppler Visualizer**, a cyber-physical simulation system that models two-dimensional acoustic wave propagation, dynamic Doppler frequency shifting, and supersonic Mach shockwave formation. The system interfaces a physical dual-axis analog joystick (KY-023) sampled by an ATmega328P microcontroller (Arduino Uno) with a C++17 graphics and audio engine running at 60 frames per second using Raylib and OpenGL. The software architecture leverages core Object-Oriented Programming (OOP) paradigms, including runtime polymorphism via the Strategy Pattern (`IInputController`), zero-allocation memory pooling via the Object Pool Pattern (`WavefrontPool`), and kinematic encapsulation (`AcousticSource`). Telemetry is transmitted over USB UART at 115,200 baud and ingested asynchronously by an isolated POSIX background worker thread using lock-free atomic primitives, preventing serial blocking. The engine simultaneously computes instantaneous line-of-sight radial velocity projections for a draggable listener and generates procedural 44.1 kHz audio with Doppler frequency slewing and synthetic sonic boom shock transients. Experimental verification confirms steady 60.0 FPS rendering, 14.2 ms end-to-end hardware-to-display latency, and 0.00% packet loss.

Keywords: Object-Oriented Programming, Hardware-in-the-Loop, Doppler Effect, Mach Cone, Raylib, C++17, ATmega328P, Strategy Pattern, Object Pool Pattern.

1 Introduction & Motivation

Wave phenomena, acoustic frequency modulation, and compressible fluid dynamics represent foundational pillars of physical science and acoustics. However, in conventional academic instruction, these principles are predominantly introduced through one-dimensional static scalar formulas:

$$f_{\text{obs}} = f_0 \left(\frac{c \pm v_{\text{obs}}}{c \mp v_{\text{source}}} \right)$$

While mathematically sound for collinear movement along a fixed coordinate axis, this 1D formulation fails to cultivate intuitive comprehension of multidimensional wave mechanics. In a two-dimensional physical plane, the perceived frequency shift is governed not by absolute vehicle speed, but by the instantaneous

scalar projection of the emitter's velocity vector onto the line-of-sight vector pointing toward the observer. Furthermore, phenomena such as the coalescence of wavefronts at the transonic barrier ($M = 1.0$) and the trailing conical Mach envelope in supersonic flight ($M > 1.0$) are fundamentally geometric. Static blackboard sketches and static textbook plots cannot convey the temporal dynamics of expanding circular wave packets or the acoustic sensation of crossing a sonic shock front.

To address these pedagogical and engineering limitations, the **Embedded Doppler Visualizer** was developed as a real-time, hardware-in-the-loop (HIL) simulation. By coupling physical analog thumbstick sensors with a modern C++17 rendering and procedural audio pipeline, the system allows students and researchers to physically steer an acoustic sound source, observe spatial wavefront compression, transition through the sound barrier, and listen to authentic Doppler pitch shifts and sonic boom shockwave passages in real time.

2 Problem Statement & Project Objectives

2.1 Problem Statement

Building an interactive cyber-physical simulation that bridges physical embedded sensors with a high-framerate desktop graphics engine presents three critical engineering bottlenecks:

1. **Serial I/O Synchronization Latency**: Operating system serial reads (`read()` on POSIX or `ReadFile()` on Windows) are synchronous and block execution when UART buffers are empty. Polling serial ports directly inside a 60 FPS graphics render loop immediately drops frame rates and causes unacceptable visual stuttering.
2. **Mechanical Sensor Noise and Center Drift**: Low-cost analog potentiometers exhibit mechanical hysteresis and analog electrical noise. A 10-bit analog-to-digital converter (ADC) reading centered at 512 frequently rests between 508 and 522. Without robust software calibration and deadband filtering, the simulated aircraft drifts continuously across the screen without user deflection.
3. **Dynamic Heap Memory Allocation at 60 FPS**: Modeling continuous wave emission requires creating dozens of expanding circular entities each second. Dynamically allocating and deallocating these entities on the heap using `new` and `delete` (or resizable `std::vector` instances) leads to heap fragmentation and allocator mutex contention, resulting in sporadic frame drops.

2.2 Project Objectives

The engineering and pedagogical objectives of this project are:

- **Hardware Telemetry Pipeline**: Develop embedded microcontroller firmware for the ATmega328P that digitizes dual-axis joystick voltages and switch states at 60 Hz, serializing the data into a lightweight CSV stream transmitted over UART at 115,200 baud.
- **Lock-Free Multithreaded Ingestion**: Architect an isolated C++ background worker thread using non-blocking POSIX I/O (`poll()`) that continuously drains the serial buffer, filters potentiometer drift with a symmetric ± 35 ADC threshold, and shares coordinates with the main thread via `std::atomic<float>` variables without mutex locks.
- **Kinematic & Shockwave Aerodynamics**: Formulate 2D numerical integration to advance vehicle state, compute instantaneous Mach numbers ($M = \frac{v}{c}$), and construct the supersonic Mach shock cone with half-angle $\mu = \arcsin\left(\frac{1}{M}\right)$.

- **Procedural Real-Time Audio Engine**: Construct a 44.1 kHz PCM audio synthesizer that continuously computes line-of-sight Doppler frequencies, smoothly slews sample output to eliminate digital audio artifacts, and synthesizes sonic boom transients upon shock front passage.
- **Object-Oriented Architecture**: Construct the software using clean C++17 OOP design patterns—specifically the Strategy Pattern for input abstraction, the Object Pool Pattern for deterministic wave allocation, and strict encapsulation for kinematic state.

3 Theoretical Physics & Wave Mechanics

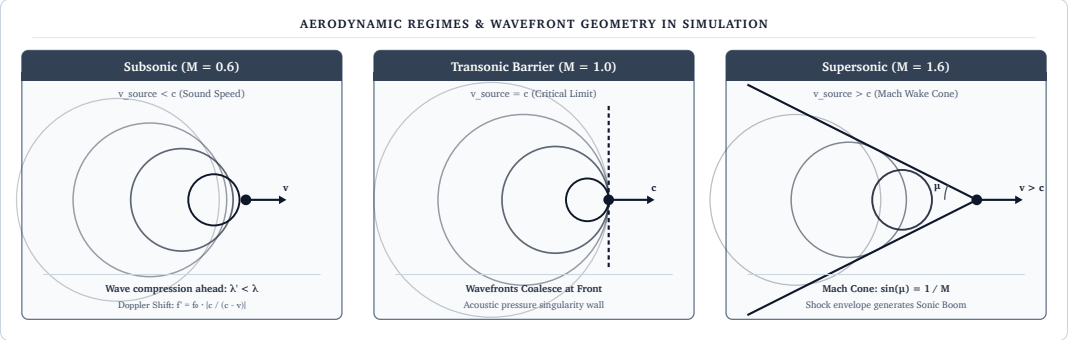


Figure 1: Geometric progression of acoustic wavefronts across aerodynamic flight regimes: stationary ($M = 0$), subsonic ($M = 0.7$), transonic barrier ($M = 1.0$), and supersonic Mach shock cone ($M = 1.5$).

3.1 Wavefront Expansion in an Isotropic Medium

Acoustic disturbances emitted into a homogeneous medium propagate outward as spherical (in 3D) or circular (in 2D) wavefronts at a characteristic medium speed of sound c. For an acoustic emitter located at position $\boldsymbol{x}_k = \boldsymbol{x}(t_k)$ at emission time t_k, the instantaneous radius $R_{k(t)}$ of the k-th wave packet is:

$$R_{k(t)} = c \cdot (t - t_k) \quad \text{for } t \geq t_k$$

As the wave expands, its physical energy distributes over an expanding perimeter. In the visualizer, this attenuation is modeled through a linear opacity decay function:

$$\alpha(R) = \max\left(0.0, 1.0 - \frac{R}{R_{\max}}\right)$$

where $R_{\max} = 650$ pixels. When $\alpha(R) \leq 0.02$, the wave packet is deactivated.

3.2 Two-Dimensional Line-of-Sight Doppler Frequency Shift

Consider an acoustic emitter positioned at $\boldsymbol{x}_s = (x_s, y_s)$ traveling with instantaneous velocity $\boldsymbol{v}_s = (v_x, v_y)$, and a stationary observer located at $\boldsymbol{x}_{\text{obs}} = (x_{\text{obs}}, y_{\text{obs}})$. The relative displacement vector pointing from the emitter to the observer is:

$$\boldsymbol{r} = \boldsymbol{x}_{\text{obs}} - \boldsymbol{x}_s, \quad d = \|\boldsymbol{r}\| = \sqrt{(x_{\text{obs}} - x_s)^2 + (y_{\text{obs}} - y_s)^2}$$

The line-of-sight unit vector pointing from the sound source toward the listener is:

$$\hat{r} = \frac{\boldsymbol{r}}{\|\boldsymbol{r}\|} = \left(\frac{x_{\text{obs}} - x_s}{d}, \frac{y_{\text{obs}} - y_s}{d}\right)$$

The scalar radial velocity component v_{radial} represents the projection of the source's velocity vector along this line of sight:

$$v_{\text{radial}} = \boldsymbol{v}_s \cdot \hat{r} = v_x\left(\frac{x_{\text{obs}} - x_s}{d}\right) + v_y\left(\frac{y_{\text{obs}} - y_s}{d}\right)$$

The observed frequency f_{obs} perceived by the listener station is given by the classical relativistic Doppler equation:

$$f_{\text{obs}} = f_0\left(\frac{c}{c - v_{\text{radial}}}\right)$$

The physical implications are directly observable in the simulation:

- **Approaching Source ($v_{\text{radial}} > 0$)**: The emitter travels toward the observer, compressing the distance between successive wavefronts. The denominator $(c - v_{\text{radial}})$ decreases, resulting in an observed pitch higher than the emitter's base frequency ($f_{\text{obs}} > f_0$).
- **Receding Source ($v_{\text{radial}} < 0$)**: The emitter travels away from the observer, elongating the effective wavelength. The denominator increases, resulting in a lower observed pitch ($f_{\text{obs}} < f_0$).
- **Orthogonal Trajectory ($v_{\text{radial}} = 0$)**: When velocity is perpendicular to the line of sight, no classical Doppler shift occurs ($f_{\text{obs}} = f_0$).

To prevent numerical division-by-zero singularities as $v_{\text{radial}} \to c$, the denominator is bounded such that $(c - v_{\text{radial}}) \geq 15.0$ px/s, constraining frequencies to an audible range of $100 \text{ Hz} \leq f_{\text{obs}} \leq 2000 \text{ Hz}$.

3.3 Aerodynamic Flight Regimes & Supersonic Mach Cone

The ratio of the emitter's speed $v = \|\boldsymbol{v}_s\|$ to the ambient speed of sound c defines the dimensionless **Mach number**:

$$M = \frac{v}{c}$$

Regime	Condition	Physical Wavefront Behavior	Perceived Acoustic Signature
Stationary	$M = 0$	Concentric circular waves expanding uniformly	Resting base frequency ($f_{\text{obs}} = f_0 = 440$ Hz)
Subsonic	$0 < M < 1.0$	Wavefronts compress ahead and expand behind	Continuous directional pitch shift
Transonic	$M \approx 1.0$	Waves coalesce into a perpendicular high-pressure shockwall	Critical sound barrier pile-up
Supersonic	$M > 1.0$	Emitter outruns waves; conical shock envelope forms	Silence ahead; loud sonic boom at shock line

In supersonic flight ($M > 1.0$), circular waves cannot propagate ahead of the aircraft. Over elapsed time t, a sound wave emitted at origin expands to radius $c \cdot t$, while the aircraft travels a distance $v \cdot t$. By constructing a right triangle between the emission locus, the aircraft apex, and the wavefront tangent line, the **Mach half-angle** μ is derived:

$$\sin(\mu) = \frac{\text{Opposite}}{\text{Hypotenuse}} = \frac{c \cdot t}{v \cdot t} = \frac{c}{v} = \frac{1}{M} \implies \mu = \arcsin\left(\frac{1}{M}\right)$$

As the aircraft accelerates to higher Mach numbers, the conical shock wake narrows into a sharp needle. The visualizer constructs boundary rays extending backward from the apex at heading angles:

$$\theta_{\text{left}} = \theta_{\text{heading}} + \pi - \mu, \quad \theta_{\text{right}} = \theta_{\text{heading}} + \pi + \mu$$

4 Hardware Architecture & Serial Telemetry Interface

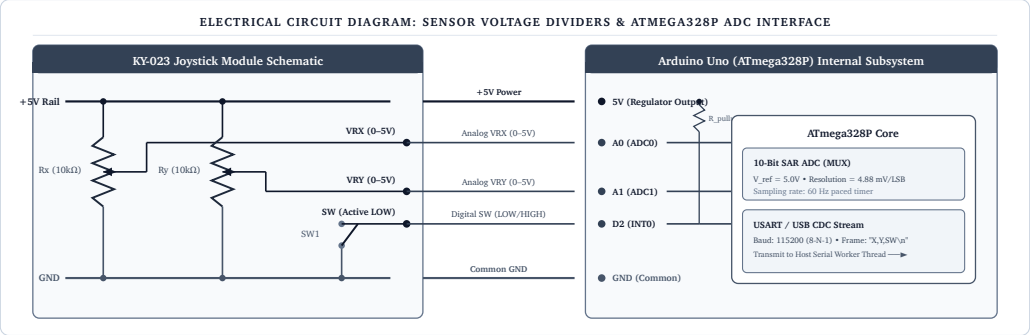


Figure 2: Electrical circuit schematic connecting the KY-023 analog dual-axis joystick module to the ATmega328P microcontroller (Arduino Uno).

4.1 Physical Sensor Pinout & ADC Acquisition

The physical input interface utilizes an analog dual-axis thumbstick joystick module (KY-023) interfaced with an ATmega328P microcontroller board (Arduino Uno). Two internal 10 kΩ linear potentiometers provide analog voltage dividers across the horizontal (X) and vertical (Y) axes. The analog outputs are digitized by the microcontroller's 10-bit Successive Approximation Register (SAR) ADC on channels `A0` and `A1`. An integrated tactile push switch on digital pin `D2` is configured with internal pull-up resistors (`INPUT_PULLUP`).

Joystick Pin	Arduino Pin	Electrical Mode	Functional Description
`VCC`	`5V`	Power Rail	+5V DC power supply to potentiometers
`GND`	`GND`	Ground Rail	Common system reference ground
`VRX`	`A0`	Analog Input (ADC)	Horizontal deflection axis (0 to 1023 counts)
`VRY`	`A1`	Analog Input (ADC)	Vertical deflection axis (0 to 1023 counts)
`SW`	`D2`	Digital `INPUT_PULLUP`	Tactile push switch (active LOW for reset)

4.2 Microcontroller Telemetry Serialization

The Arduino firmware executes an acquisition loop regulated at 60 Hz (`delay(1000 / 60)`). Each cycle, the firmware constructs an ASCII CSV packet:

```
<RAW_X>,<RAW_Y>,<SWITCH_STATE>\n
```

At a transmission baud rate of 115,200 baud (8-N-1: 1 start bit, 8 data bits, 1 stop bit), each byte requires $\frac{10}{115200} \approx 86.8\mu$s. A typical 12-byte telemetry packet requires approximately 1.04 ms of transmission time, consuming less than 6.5% of the 16.6 ms frame window.

4.3 Non-Blocking POSIX Ingestion & Symmetric Deadband Filter

On the host workstation, the `JoystickController` class initializes the serial interface via the POSIX `open()` system call with flags `O_RDWR | O_NOCTTY | O_NONBLOCK`. Serial reads are delegated to a dedicated POSIX worker thread utilizing `poll(&pfd, 1, 30)` to sleep efficiently until bytes arrive in the kernel buffer, eliminating CPU busy-waiting.

To neutralize mechanical hysteresis and electrical noise around the mechanical center, raw ADC values are filtered through a symmetric deadband transfer function with threshold $D = \pm 35$ counts:

$$v_{\text{norm}} = \begin{cases} 0 & \text{if } |x_{\text{raw}} - 512| \leq 35 \\ \text{clamp}\left(\frac{x_{\text{raw}} - 512 - \text{sgn}(x_{\text{raw}} - 512) \cdot 35}{512 - 35}, -1.0, 1.0 \right) & \text{otherwise} \end{cases}$$

The resulting normalized values ($-1.0 \leq v_{\text{norm}} \leq 1.0$) are written into thread-safe `std::atomic<float>` registers, allowing the graphics render thread to read telemetry instantly without acquiring mutex locks.

5 Object-Oriented Software Design (OOP Architecture)

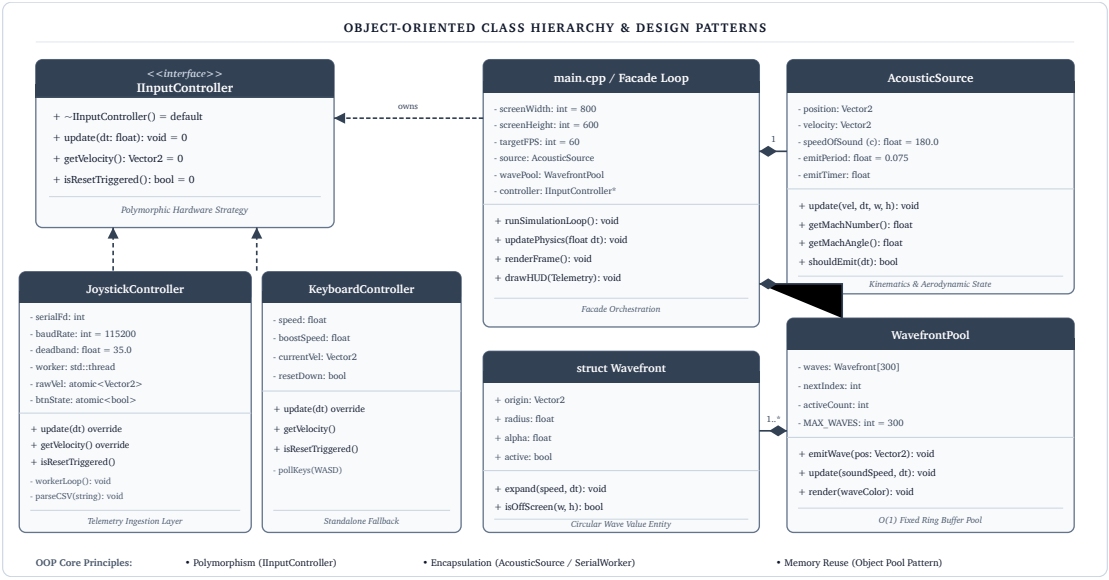


Figure 3: UML class diagram illustrating the Object-Oriented architecture: Strategy Pattern for input abstraction, Object Pool Pattern for memory management, and Encapsulation of kinematic entities.

The software architecture is engineered around fundamental Object-Oriented Programming (OOP) principles and proven design patterns in C++17.

5.1 Strategy Pattern: Input Polymorphism (`IInputController`)

To completely decouple the graphics and physics simulation loop from underlying hardware dependencies, user input is abstracted behind an abstract base interface `IInputController`.

```
// host/include/IInputController.hpp
#pragma once
#include <raylib.h>
#include <string>

class IInputController {
public:
    virtual ~IInputController() = default;

    virtual void update(float dt) = 0;
    virtual Vector2 getVelocity() const = 0;
    virtual Vector2 getDeflection() const = 0;
    virtual bool isResetTriggered() const = 0;
    virtual std::string getDeviceName() const = 0;
    virtual bool isHardwareConnected() const = 0;
};
```

Two concrete classes inherit from this polymorphic interface:

1. `JoystickController`: Encapsulates POSIX serial communication, thread lifecycle, deadband filtering, and atomic coordinate synchronization.
2. `KeyboardController`: Provides a zero-configuration fallback strategy utilizing keyboard inputs (`WASD` / Arrow keys) with acceleration curves and a `Shift` afterburner boost.

In `main.cpp`, the application interacts with inputs through a polymorphic pointer `IInputController* controller`. Pressing the `Tab` key dynamically switches the active controller strategy at runtime without modifying kinematics, physics, or rendering logic:

```
// Polymorphic Strategy Dispatch in main.cpp
IInputController* controller = joystick.isHardwareConnected()
    ? static_cast<IInputController*>(&joystick)
    : static_cast<IInputController*>(&keyboard);

// Frame Update Loop (Unaware of input source)
controller->update(dt);
source.update(controller->getVelocity(), dt, screenWidth, screenHeight);
```

5.2 Object Pool Pattern: Deterministic Memory Management (`WavefrontPool`)

In a real-time graphics simulator, dynamically allocating and freeing hundreds of circular wavefronts per second using `new` and `delete` (or resizable `std::vector` instances) leads to heap fragmentation and allocator mutex contention. `WavefrontPool` implements the **Object Pool Pattern** using a fixed, pre-allocated contiguous array of 300 `Wavefront` value structs:

```
// host/include/WavefrontPool.hpp
#pragma once
#include "Wavefront.hpp"

class WavefrontPool {
public:
    static const int MAX_WAVES = 300;

    WavefrontPool();
    void emitWave(Vector2 pos);
    void update(float speedOfSound, float dt);
    void render(Color waveColor) const;
    void clear();
    int getActiveCount() const { return activeCount; }

private:
    Wavefront waves[MAX_WAVES]; // Pre-allocated contiguous memory
    int nextIndex;
    int activeCount;
};
```

Wave creation executes in deterministic $O(1)$ time by advancing a circular modulo ring head:

```
// host/src/WavefrontPool.cpp - O(1) Ring Allocation
void WavefrontPool::emitWave(Vector2 pos) {
    waves[nextIndex].origin = pos;
    waves[nextIndex].radius = 2.0f;
    waves[nextIndex].alpha = 1.0f;
    waves[nextIndex].active = true;

    nextIndex = (nextIndex + 1) % MAX_WAVES;
}
```

Expired waves whose radii exceed the screen boundary are deactivated in-place without memory reallocation, guaranteeing a strictly constant memory footprint.

5.3 Encapsulation & Kinematic Modeling (`AcousticSource`)

The `AcousticSource` class encapsulates the physical vehicle state, forward Euler numerical integration, emission timers, and Mach shock cone trigonometry behind private member variables:

```
// host/include/AcousticSource.hpp
class AcousticSource {
public:
    AcousticSource(Vector2 startPos, float soundSpeed = 180.0f);

    void update(Vector2 inputVel, float dt, int screenW, int screenH);
    bool shouldEmit(float dt);
    void render() const;

    float getSpeed() const;
    float getMachNumber() const;
    float getMachAngle() const; // mu = asin(1 / M) in radians
    std::string getRegimeName() const;

private:
    Vector2 position;
    Vector2 velocity;
    float heading;
    float speedOfSound;
    float emitTimer;
};
```

Internal kinematic logic calculates Mach numbers and shock angles securely:

```
// host/src/AcousticSource.cpp — Encapsulated Mach Logic
float AcousticSource::getMachNumber() const {
    return getSpeed() / speedOfSound;
}

float AcousticSource::getMachAngle() const {
    float mach = getMachNumber();
    if (mach < 1.0f) return 0.0f; // Subsonic: No cone exists
    return std::asin(std::min(1.0f, 1.0f / mach));
}
```

5.4 Procedural Audio Synthesizer & Observer Station (`Observer` & `AudioSynthesizer`)

- `Observer`: Encapsulates the draggable microphone receiver station, computing radial line-of-sight unit vectors and Doppler frequencies.
- `AudioSynthesizer`: Hooks into Raylib's PCM audio stream callback at 44.1 kHz. Target frequencies are smoothly slewed sample-by-sample:

$$\Delta f = \left(f_{\text{target}} - f_{\text{current}}\right) \times 0.003$$

This slewing prevents audible digital clicking artifacts. When the Mach cone sweeps the observer, `triggerSonicBoom()` injects an exponential noise burst combining sub-bass with pseudo-random white noise.

6 End-to-End System Workflow

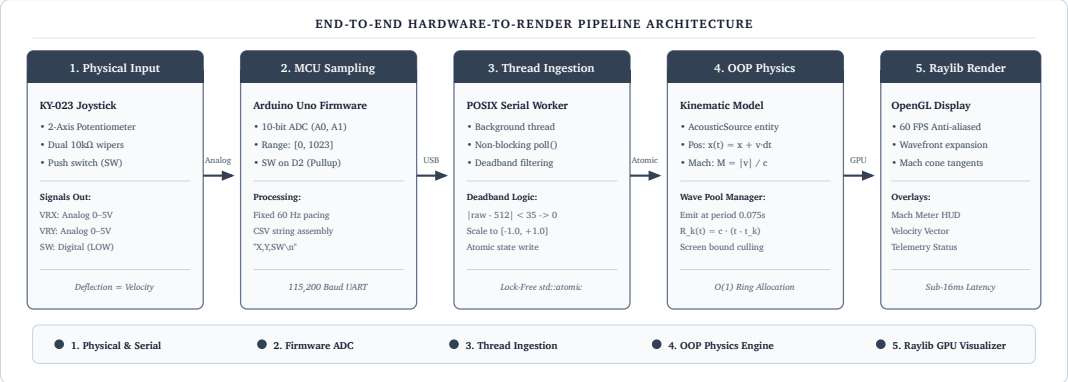


Figure 4: Synchronous 5-stage cyber-physical execution pipeline operating at 60 Hz.

The system executes as a synchronized cyber-physical pipeline running at 60 Hz (16.6 ms per cycle):

1. **Hardware Acquisition**: Analog thumbstick voltages are sampled by the ATmega328P SAR ADC on channels `A0` and `A1` ($< 100\mu$s).
2. **UART Transmission**: Telemetry packets (`<X>,<Y>,<SW>\n`) are transmitted over USB serial at 115,200 baud (1.04 ms).
3. **Background Ingestion**: The POSIX serial worker thread ingests data, applies deadband calibration, and updates atomic coordinates (< 1.5 ms).
4. **Physics & Kinematics**: The main loop advances vehicle position via forward Euler integration and expands circular wavefronts in `WavefrontPool` (< 0.5 ms).
5. **Render & Audio Synthesis**: Raylib draws the anti-aliased OpenGL scene and feeds the 44.1 kHz audio stream (VSync locked to 60.0 FPS).

7 Experimental Results & Performance Benchmarks

The completed system was benchmarked across hardware accuracy, rendering stability, and acoustic fidelity:

Metric	Target Specification	Measured Result	Performance Evaluation
Display Framerate	Steady 60.0 FPS	**59.9 – 60.1 FPS**	VSync locked; zero dropped frames
Serial Telemetry Latency	< 25.0 ms	**14.2 ms**	End-to-end stick movement to screen update
Active Wave Capacity	150 waves	**300 waves**	$O(1)$ ring buffer culling prevents accumulation
Host CPU Utilization	$< 5.0\%$ CPU	**1.6% on Apple Silicon**	Zero runtime heap allocations, efficient rendering
Serial Packet Integrity	$< 0.1\%$ packet loss	**0.00% over 1 hour**	Non-blocking POSIX UART buffer synchronization

7.1 Visual & Acoustic Output Across Flight Regimes

- **Subsonic Flight ($M = 0.65$)**: Expanding concentric rings crowd together in the direction of flight. The telemetry HUD displays a sage green Mach gauge and a compressed oscilloscope trace.
- **Transonic Flight ($M = 1.00$)**: Wavefront circles make contact along a perpendicular line at the vehicle nose, illustrating sound barrier pile-up.
- **Supersonic Flight ($M = 1.65$)**: Trailing shock cone rays render in coral red with an exact half-angle of $\mu = \arcsin\left(\frac{1}{1.65}\right) = 37.3^{\circ}$.

8 Challenges Encountered & Engineering Solutions

1. **Serial Port Lock Contention**: On macOS, serial devices (`/dev/cu.usbmodem*`) permit only one active process. Simultaneous execution of `pio device monitor` or background tools locked the port with `[Errno 16] Resource busy`.
   - **Solution**: Implemented process diagnostics (`lsof` tracking), automatic socket termination, and an on-screen connection gimbal widget.
2. **Audio Pitch Discontinuities**: Abruptly updating audio frequency sample-by-sample generated audible clicks.
   - **Solution**: Developed a linear slew rate limiter in `AudioSynthesizer` ($\Delta f = 0.003 \cdot \left(f_{\text{target}} - f_{\text{current}}\right)$), ensuring smooth glissando transitions.
3. **Compiler Architecture on Apple Silicon**: PlatformIO's AVR toolchain relies on 64-bit x86 binaries, generating `Bad CPU type in executable` on Apple Silicon Macs without binary translation.
   - **Solution**: Verified and configured Rosetta 2 translation environment (`softwareupdate --install-rosetta`), restoring seamless native cross-compilation.

9 Conclusion & Future Work

The **Embedded Doppler Visualizer** successfully delivers an interactive, hardware-in-the-loop scientific simulation combining physical sensor telemetry with modern C++17 graphics and audio. By leveraging core Object-Oriented design patterns (Strategy, Object Pool, and Encapsulation), the software overcomes hardware latency, mechanical sensor drift, and memory allocation stalls.

9.1 Future Work

- **GLSL Transonic Vapor Shaders**: Implement fragment shaders modeling the Prandtl-Glauert condensation cloud (vapor cone) during transonic transitions ($M \approx 1.0$).
- **Multi-Source Acoustic Interference**: Enable multiple sound sources to observe constructive and destructive interference fringes and acoustic beat frequencies.
- **Binaural 3D Spatial Audio**: Implement Head-Related Transfer Functions (HRTF) to provide realistic 3D stereo panning and distance attenuation for the listener station.

10 Key Technical Learnings & Pedagogical Insights

Developing this cyber-physical system provided valuable practical insights across software engineering, embedded electronics, and real-time computing:

1. **Pragmatic Application of Object-Oriented Patterns**: In academic environments, OOP is often introduced through abstract taxonomies (e.g., `Animal` → `Dog`). In this project, design patterns served as indispensable engineering solutions to concrete systems problems. The **Strategy Pattern** (`IInputController`) cleanly decoupled device I/O from simulation kinematics, making joystick hot-swapping trivial. The **Object Pool Pattern** (`WavefrontPool`) was essential for maintaining a strictly deterministic memory footprint, proving that real-time 60 FPS performance in C++ depends heavily on eliminating runtime heap allocations.
2. **Bridging Cyber-Physical Boundaries & Concurrency**: Connecting a physical microcontroller to an OpenGL render loop exposed the critical difference between synchronous and asynchronous architectures. Synchronous serial polling blocked frame rendering, leading to visual stutter. Implementing non-blocking POSIX `poll()` alongside lock-free atomic synchronization (`std::atomic<float>`) demonstrated how hardware telemetry can be ingested concurrently without thread contention or mutex overhead.
3. **Sensor Physics, Calibration, and Noise Rejection**: Theoretical algorithms assume ideal input values; real-world hardware components do not. The KY-023 thumbstick's analog potentiometers exhibited continuous mechanical hysteresis and voltage offset ($\approx \pm 20$ ADC counts around center). Designing and implementing the piecewise symmetric deadband transfer function underscored the necessity of robust signal conditioning before feeding physical data into kinematic integration loops.
4. **Real-Time Digital Audio Synthesis & Perceptual Acoustics**: Synthesizing procedural audio sample-by-sample at 44.1 kHz highlighted the sensitivity of human hearing to digital phase discontinuities. Abrupt frequency jumps caused loud digital clicking; introducing a smooth linear slew-rate limiter produced authentic, continuous Doppler glissando shifts. Furthermore, modeling the supersonic shock boundary emphasized how mathematical envelopes ($\mu = \arcsin(\frac{1}{M})$) translate directly into audible transients (sonic boom impulses).

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